The Internet

Elliot Attipoe

Internet Usage Stats

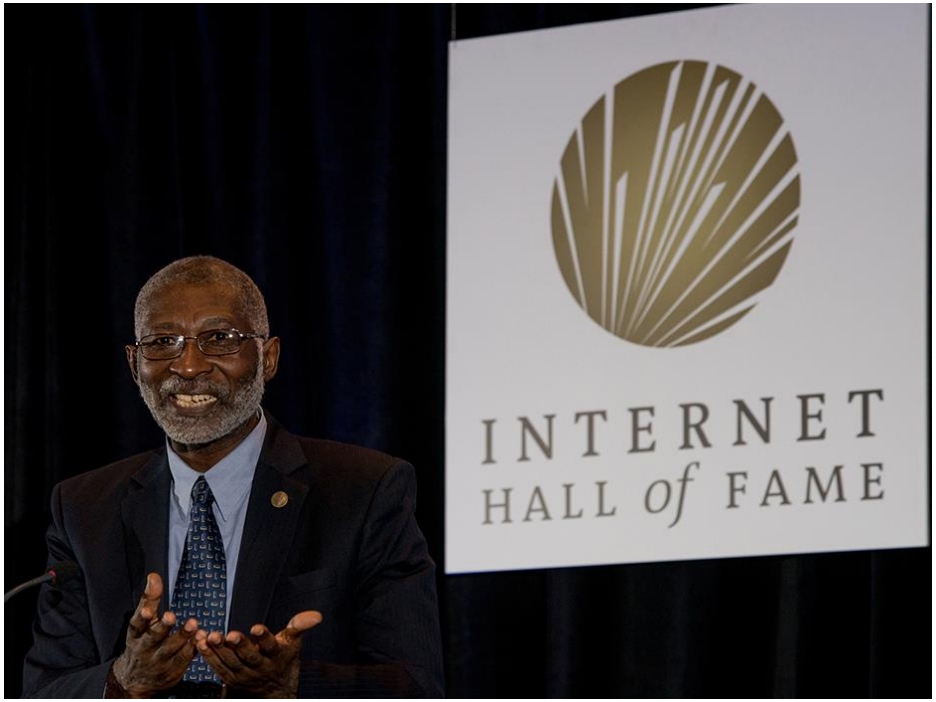
WORLD INTERNET USAGE AND POPULATION STATISTICS 2019 Mid-Year Estimates						
World Regions	Population (2019 Est.)	Population % of World	Internet Users 30 June 2019	Penetration Rate (% Pop.)	Growth 2000-2019	Internet World %
Africa	1,320,038,716	17.1 %	522,809,480	39.6 %	11,481 %	11.5 %
Asia	4,241,972,790	55.0 %	2,300,469,859	54.2 %	1,913 %	50.7 %
Europe	829,173,007	10.7 %	727,559,682	87.7 %	592 %	16.0 %
Latin America / Caribbean	658,345,826	8.5 %	453,702,292	68.9 %	2,411 %	10.0 %
Middle East	258,356,867	3.3 %	175,502,589	67.9 %	5,243 %	3.9 %
North America	366,496,802	4.7 %	327,568,628	89.4 %	203 %	7.2 %
Oceania / Australia	41,839,201	0.5 %	28,636,278	68.4 %	276 %	0.6 %
WORLD TOTAL	7,716,223,209	100.0 %	4,536,248,808	58.8 %	1,157 %	100.0 %

AFRICA 2019 POPULATION AND INTERNET USERS STATISTICS

AFRICA	Population (2019 Est.)	Internet Users 31-Dec-2000	Internet Users 30-June-2019	Penetration (% Population)	Internet Growth % 2000 - 2019	Facebook subscribers 31-Dec-2018
Algeria	42,679,018	50,000	25,428,159	59.6 %	50,756 %	19,000,000
Mali	19,689,140	18,800	12,480,176	63.4 %	66,284 %	1,500,000
Mauritania	4,661,149	5,000	969,519	20.8 %	19,290 %	770,000
Mauritius	1,271,368	87,000	803,896	63.2 %	824 %	700,000
Burkina Faso	20,321,560	10,000	3,704,265	18.2 %	36,942 %	840,000
Cote d'Ivoire	25,531,083	40,000	11,953,653	46.8 %	29,784 %	3,800,000
Djibouti	985,690	1,400	548,832	55.7 %	39,102 %	180,000
Egypt	101,168,745	450,000	49,231,493	48.7 %	10,840 %	35,000,000
Namibia	2,641,996	30,000	1,347,418	51.0 %	4,391 %	570,000
Niger	23,176,691	5,000	2,368,658	10.2 %	47,273 %	440,000
Nigeria	200,962,417	200,000	123,486,615	61.4 %	61,643 %	17,000,000
Ethiopia	110,135,635	10,000	20,507,255	18.6 %	204,972 %	4,500,000
Gabon	2,109,099	15,000	1,307,641	62.0 %	8,617 %	620,000
Gambia	2,228,075	4,000	442,050	19.8 %	10,951 %	310,000
Ghana	30,096,970	30,000	11,737,818	39.0 %	39,026 %	4,900,000
Guinea	13,398,180	8,000	2,411,672	18.0 %	30,046 %	1,500,000
Guinea-Bissau	1,953,723	1,500	150,000	7.7 %	9,900 %	110,000
Kenya	52,214,791	200,000	46,870,422	89.8 %	23,335 %	7,000,000

History of Internet in Ghana

- Ghana is known to be the first country in West Africa to be connected to the Internet in **1994.**
- Prof. Nii Narku Quaynor is attributed to have brought internet to Ghana.



Ghana Internet Usage

Year	Internet Users**	Penetration (% of Pop)	Total Population	Non-Users (Internetless)	1Y User Change	1Y User Change	Population Change
2016*	**7,958,675**	28.4 %	28,033,375	20,074,700	14 %	976,984	2.27 %
2015*	**6,981,691**	25.5 %	27,409,893	20,428,202	37.9 %	1,919,024	2.33 %
2014	**5,062,667**	18.9 %	26,786,598	21,723,931	57.3 %	1,844,442	2.38 %
2013	**3,218,225**	12.3 %	26,164,432	22,946,207	18.9 %	510,501	2.43 %
2012	**2,707,724**	10.6 %	25,544,565	22,836,841	20.7 %	464,159	2.47 %
2011	**2,243,565**	9 %	24,928,503	22,684,938	18.3 %	346,782	2.51 %
2010	**1,896,783**	7.8 %	24,317,734	22,420,951	47 %	606,787	2.55 %
2009	**1,289,996**	5.4 %	23,713,164	22,423,168	30.7 %	302,946	2.58 %
2008	**987,050**	4.3 %	23,115,919	22,128,869	13.8 %	119,720	2.61 %

Mobile Penetration Rate

Ghana: Mobile phone penetration soars to 128%

By **Dasmani Laary**
Posted on Wednesday, 9 March 2016 12:46

General News of Tuesday, 20 March 2018

29 million Ghanaians use 34 million mobile phones

Ghana's mobile phone subscribers to hit 40m by 2020 - Report

Source: Shiela Tamakloe | JoyBusiness

Date: 19-03-2018 Time: 04:03:10:pm

Networking Fundamentals

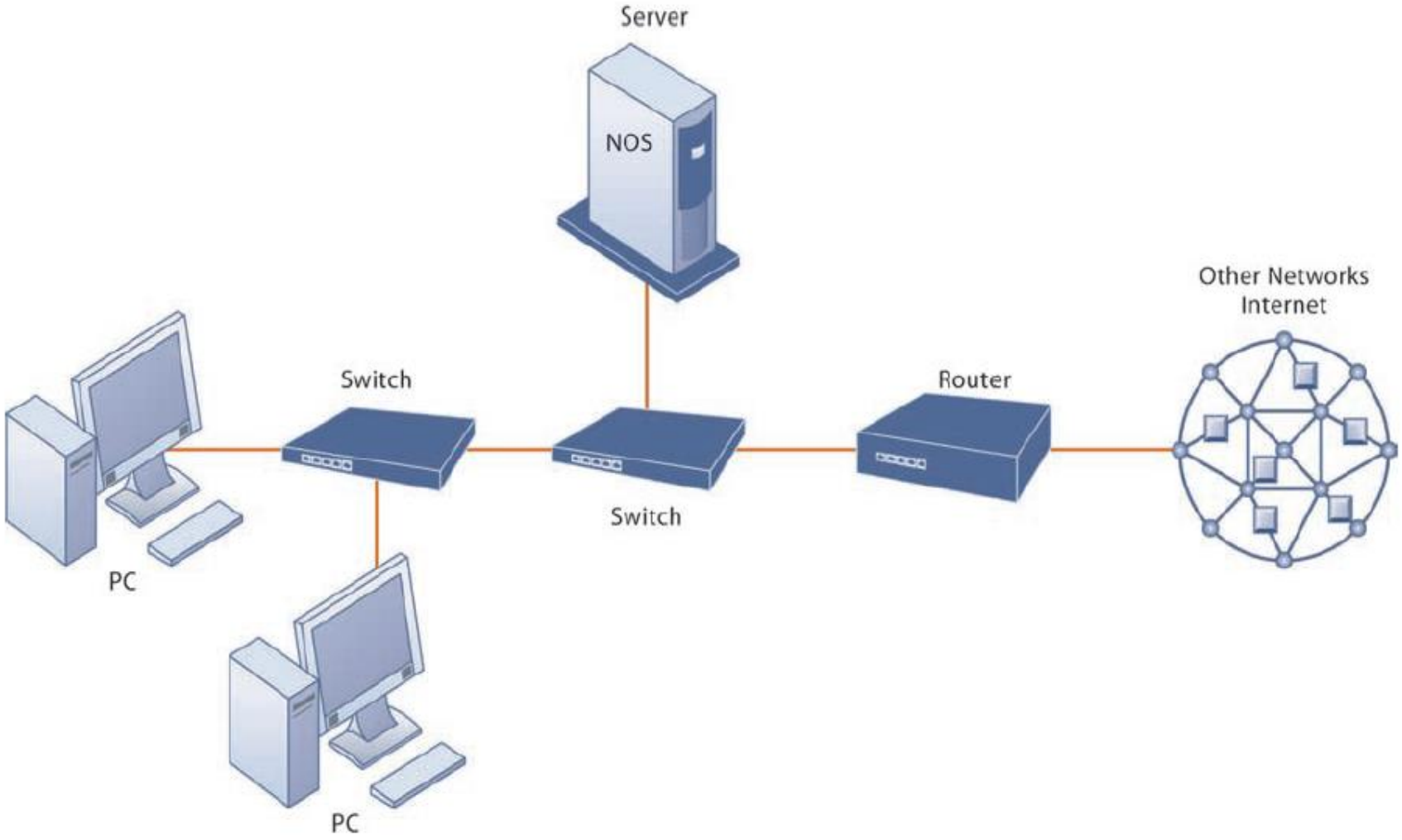
- Computer network:
  - Two or more computers connected together

- Benefits of a network:
  - Sharing resources
  - Transferring files

Components of computer network

1. Server
2. Transmission media
3. Network Interface Card
4. Switch/Hub
5. Router
6. PC/Client/work station/node

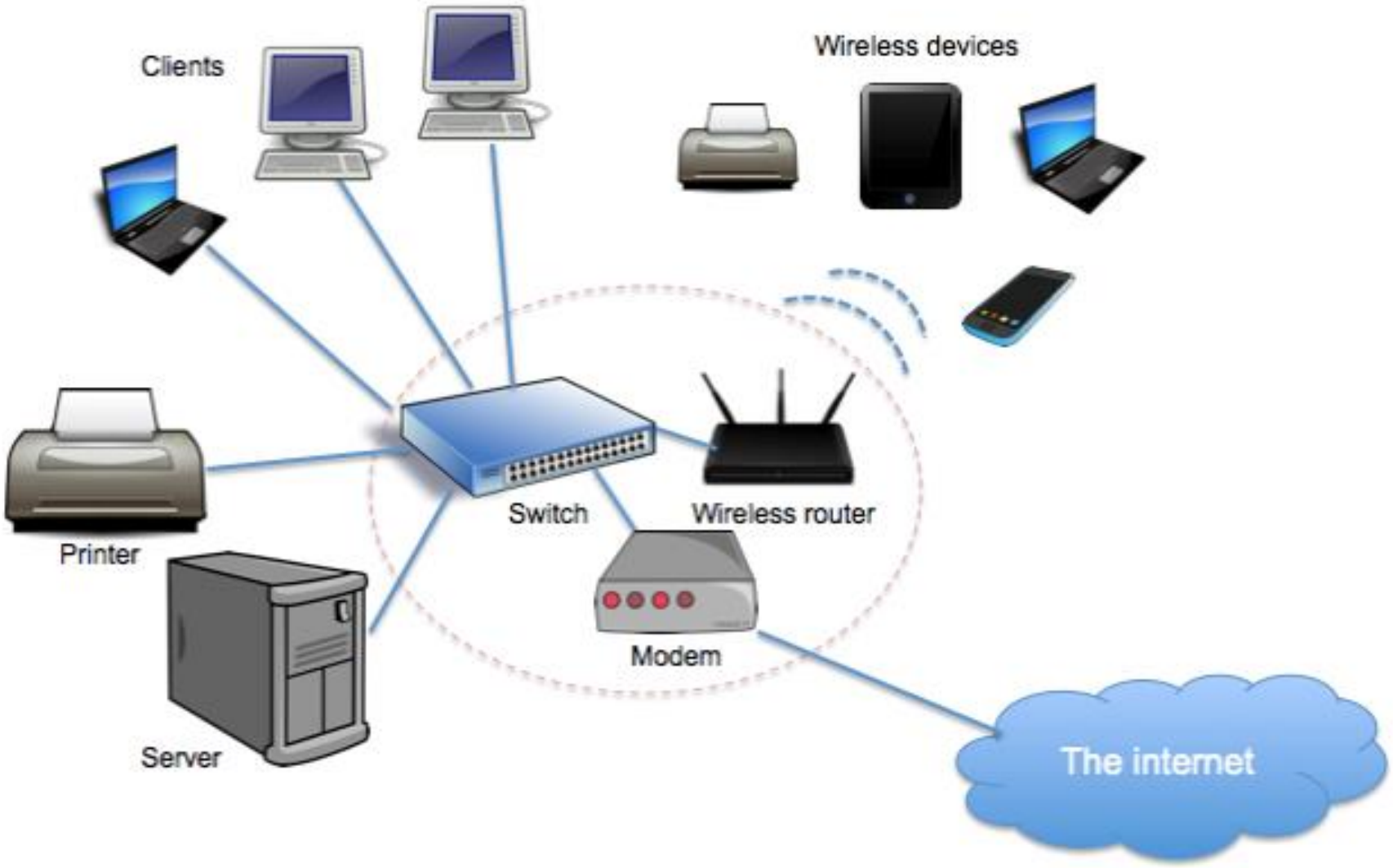
Computer Network



Wireless network



Corporate Network



Servers



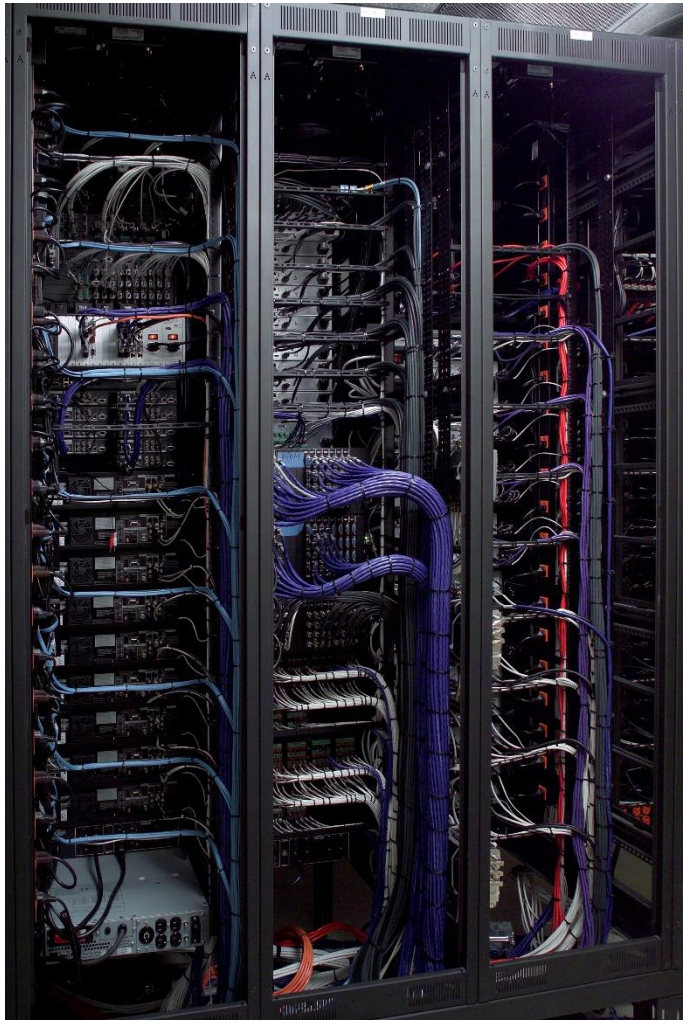


Servers

- Servers are computers that hold shared files, programs, and the network operating system.
- Servers provide access to network resources to all the users of the network.
- There are many different kinds of servers, and one server can provide several functions.
- For example, there are file servers, print servers, mail servers, communication servers, database servers, fax servers and web servers, to name a few.

Rackmount Servers

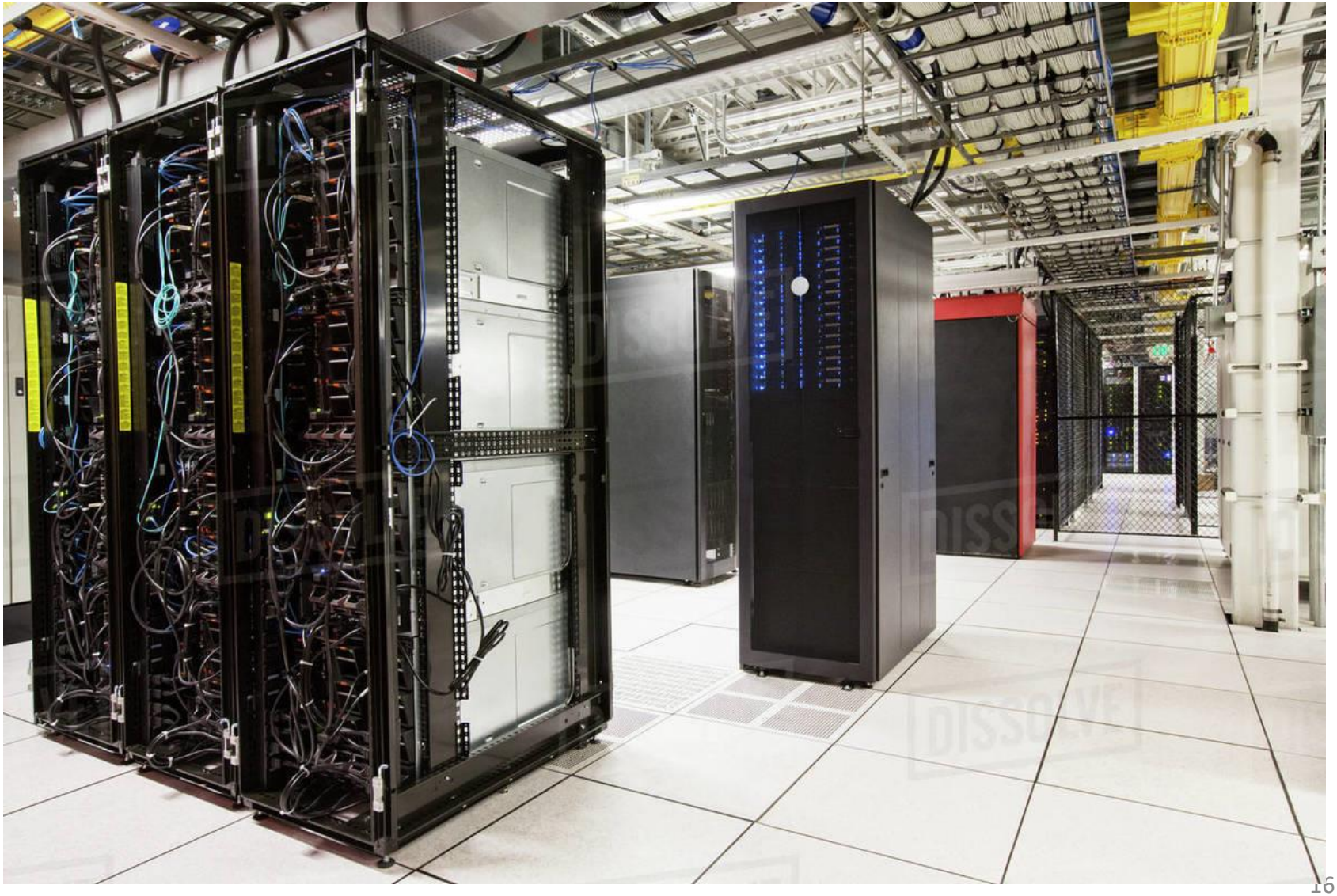




Data Center



Data Center



Data center

Google Unveils $13 Billion in New Data Center Construction for 2019

BY **RICH MILLER** - FEBRUARY 13, 2019 — 3 COMMENTS

Facebook to Build $1 Billion Data Center in Singapore

By Renae Reints September 6, 2018

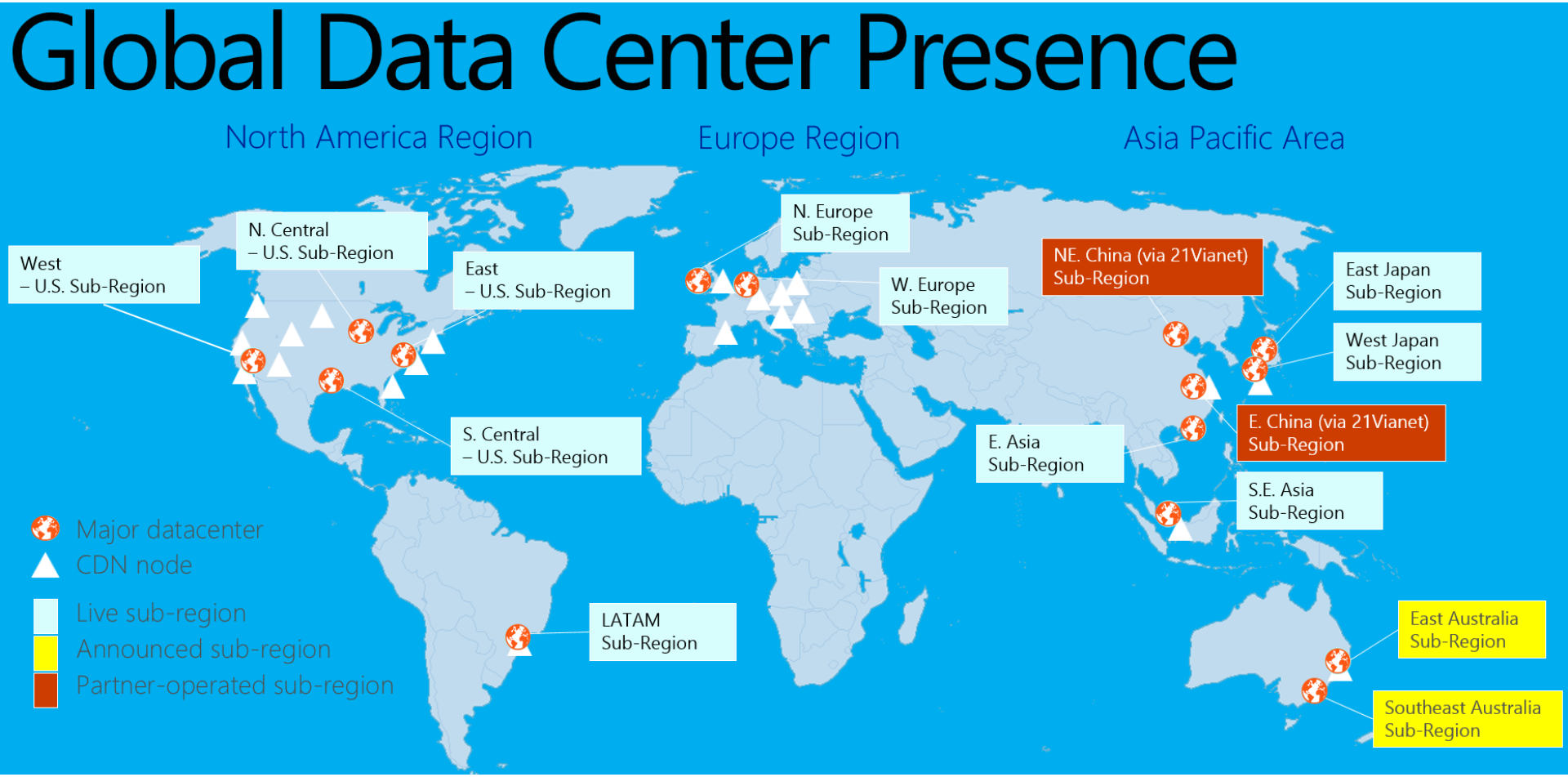
Data Centre Hardware And Software Spend Hit $150bn In 2018 Due To Cloud – Report

By Abigail Opiah | **PUBLISHED:** 10:03, 2 April, 2019 | **UPDATED:** 11:50, 2 April, 2019

Google Data Center Locations



Microsoft Azure Data Centers



Azure regions

Azure has more global regions than any other cloud provider—offering the scale needed to bring applications closer to users around the world, preserving data residency and offering comprehensive compliance and resiliency options for customers.

54 regions worldwide

140 available in 140 countries



Network Interface Card





RJ45 + Network Interface Card





Network Interface Card

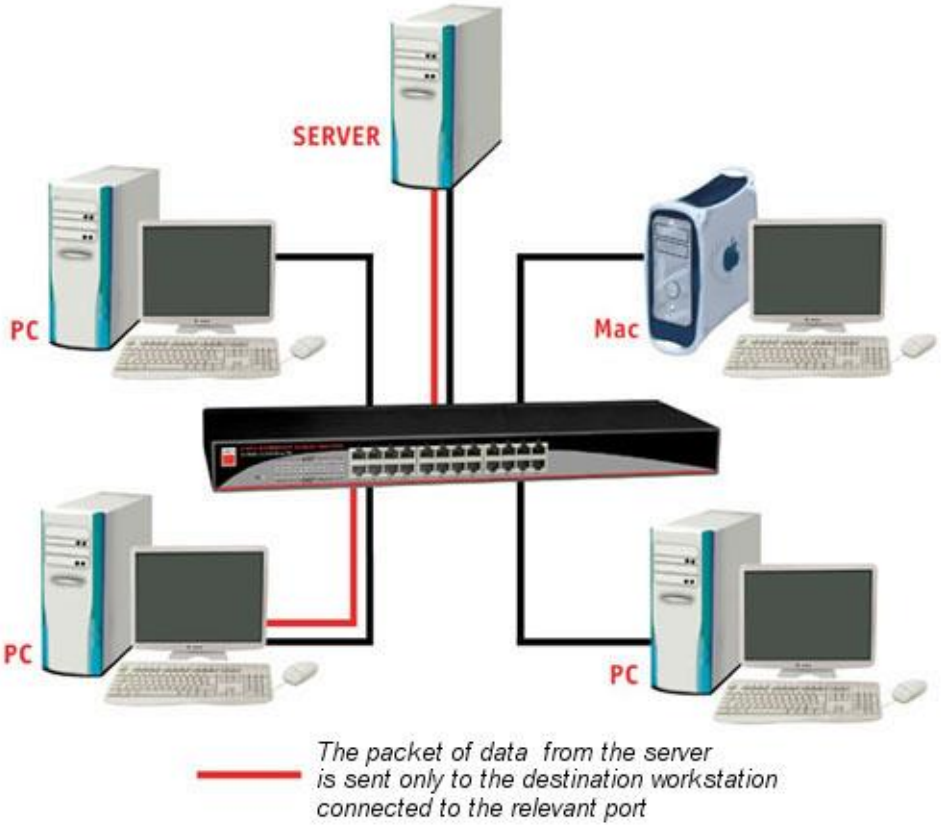
- Network Interface Card is a communications device that enables a computer or device that does not have built-in networking capability to access a network.

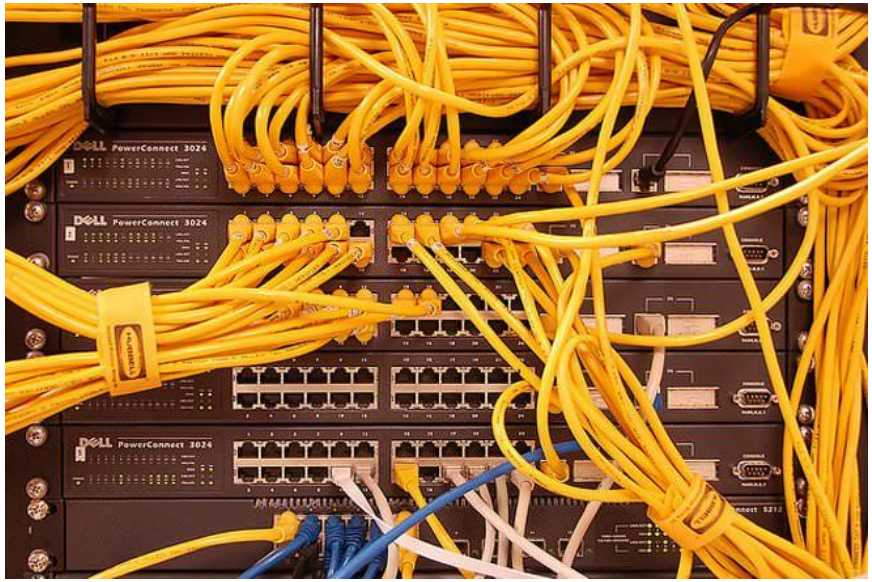
- The network card coordinates the transmission and receipt of data, instructions, and information to and from the computer or device containing the network card.

Hub/Switch



Computers connected to a switch





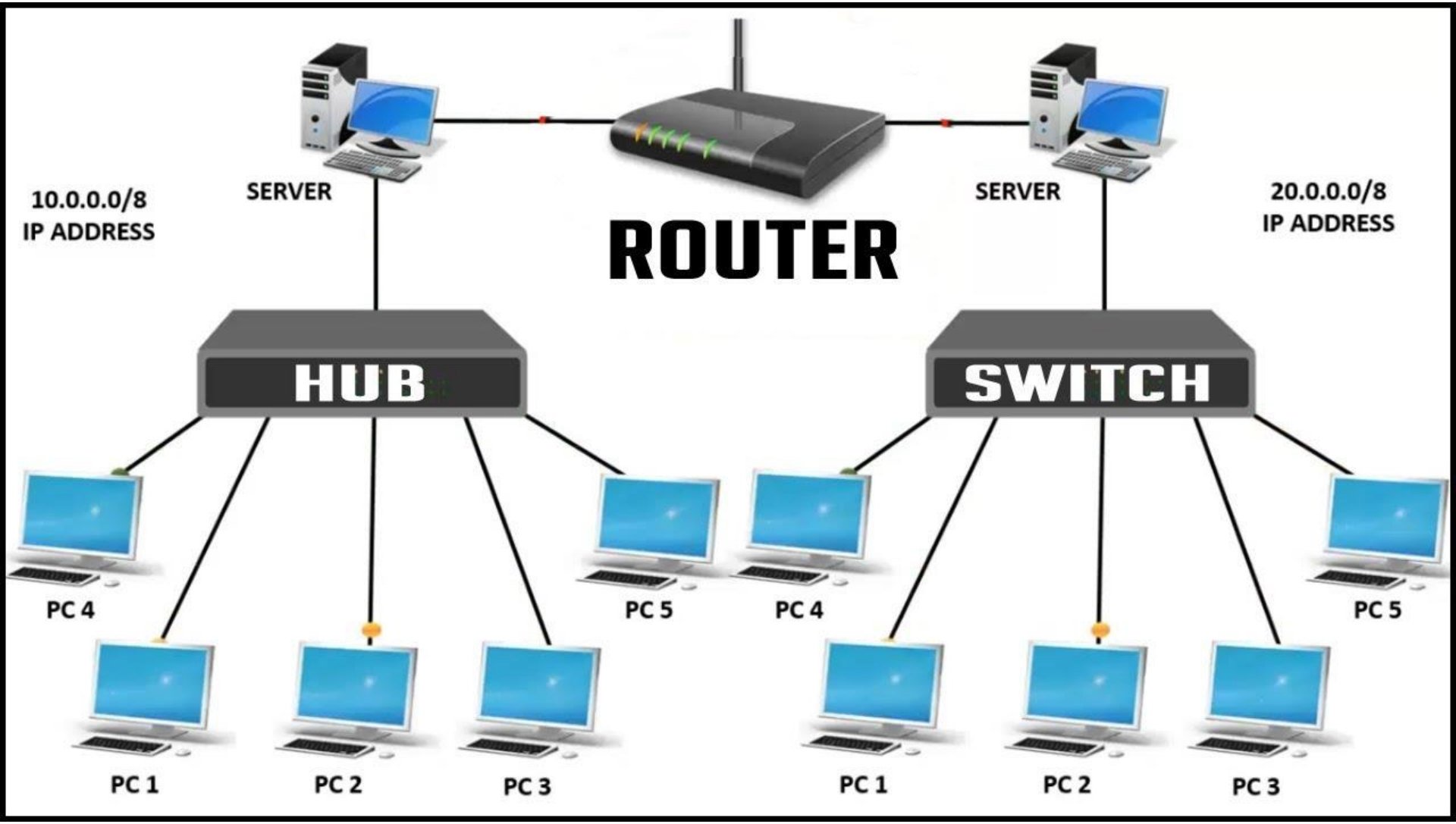
Computer Lab



Hub and switches

- A hub or switch is a device that **provides a central point for cables in a network.**
- Larger networks typically use a hub, while smaller networks use a switch.
- Hubs are very simple devices that connect network components, sending a packet of data to all other connected devices.
- A switch has more intelligence than a hub and can filter and forward data to a specified destination on the network.

Router


SERVER
SERVER
10.0.0.0/8
IP ADDRESS
20.0.0.0/8
IP ADDRESS
ROUTER
HUB
SWITCH
PC 4
PC 5
PC 4
PC 5
PC 1
PC 2
PC 3
PC 1
PC 2
PC 3

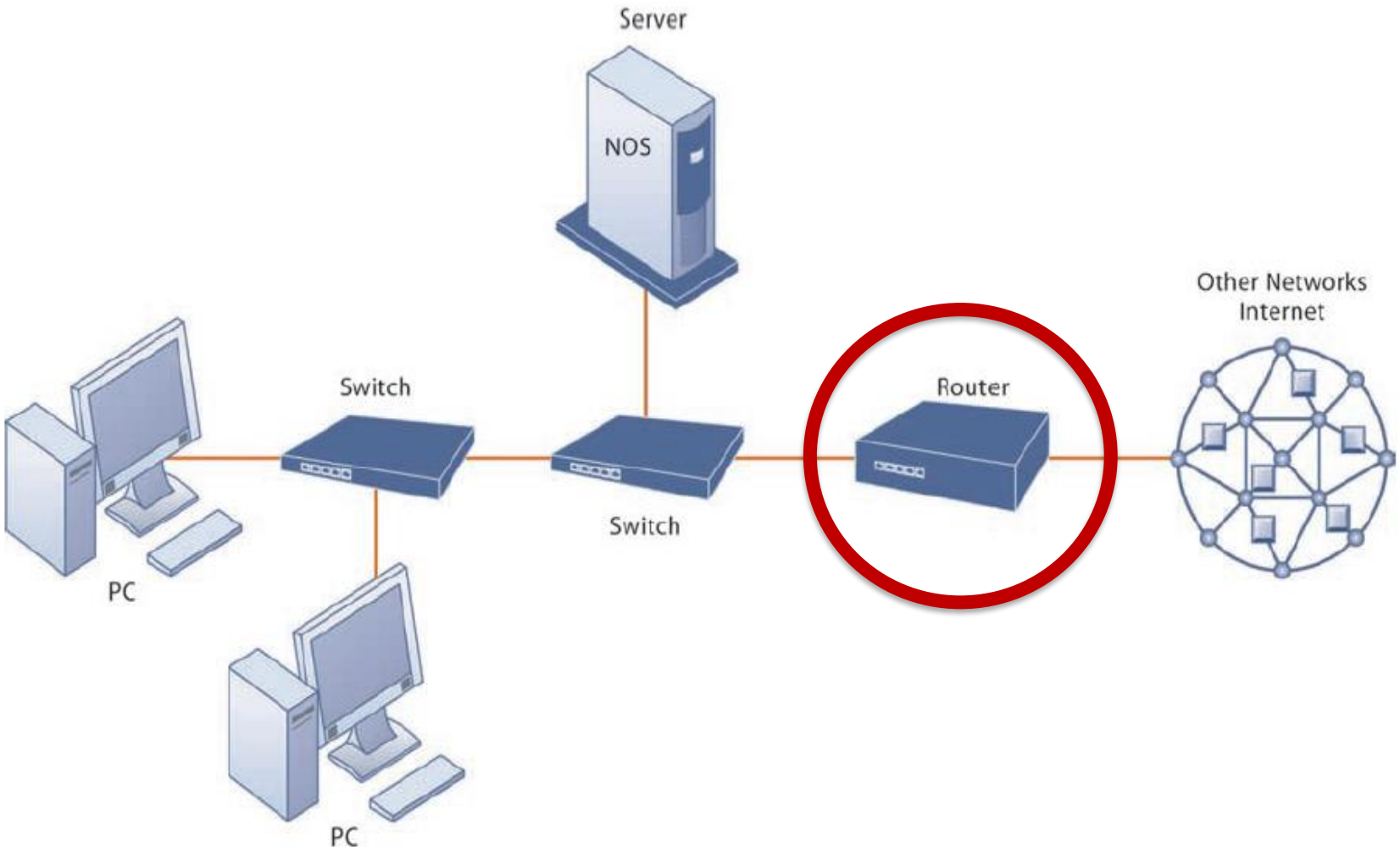
Router


wireless access point
is center of
wireless network
wired network
Internet
high-speed Internet connection

Router

- A router is a communications device that **connects multiple computers or other routers together** and transmits data to its correct destination on a network.
- A router forwards data packets between computer networks.
- Routers perform the traffic directing functions on the Internet.
- A router can be used on a network of any size.

Computer network



Transmission Media

- Transmission media consist of materials or substances capable of carrying one or more communications signals.

- When you send data from a computer or mobile device, the signal that carries the data
may travel over various transmission media.

- Classification
  - Physical Transmission Media
  - Wireless Transmission Media

Physical Transmission Media

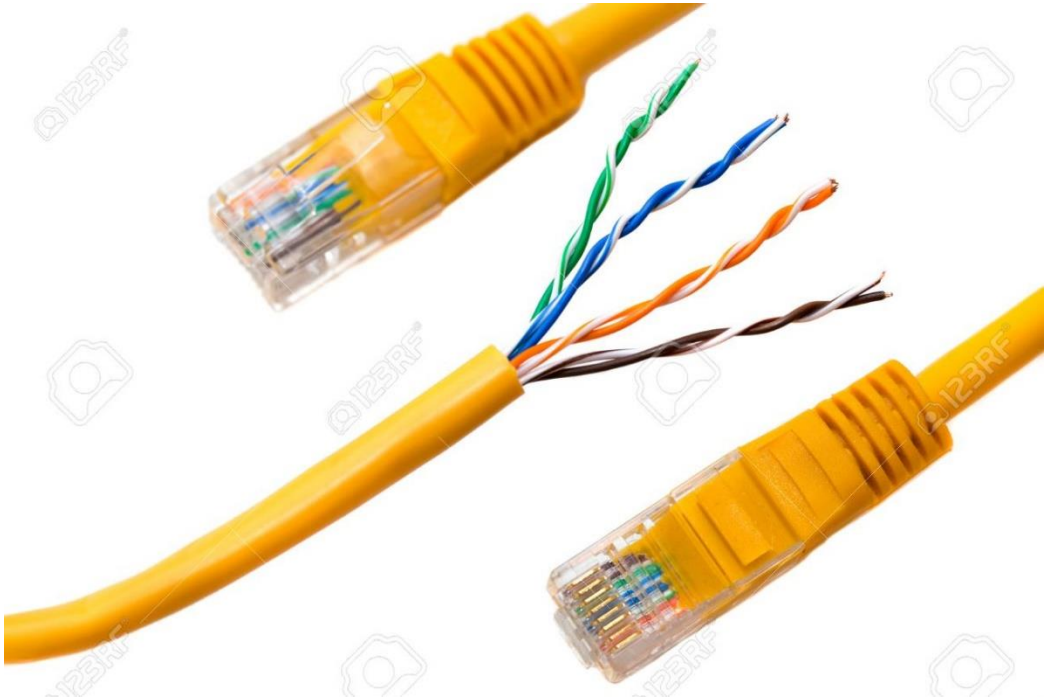
- Physical transmission media use wire, cable, and other tangible materials to send communications signals.

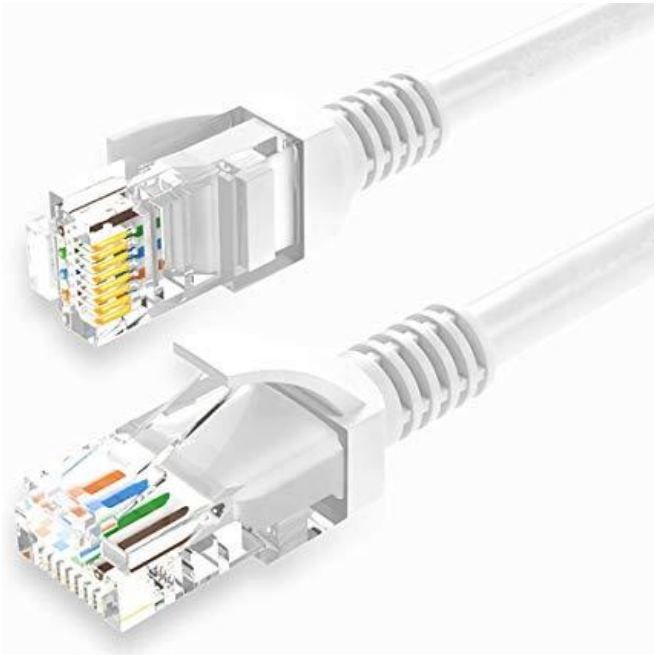
- These wires and cables typically are used within, underground, or between buildings.

Table 10-3 Transfer Rates for Physical Transmission Media Used in LANs

Type of Cable and LAN	Maximum Transfer Rate
Twisted-Pair Cable	
• 10Base-T (Ethernet)	10 Mbps
• 100Base-T (Fast Ethernet)	100 Mbps
• 1000Base-T (Gigabit Ethernet)	1 Gbps
• Token ring	4 Mbps to 16 Mbps
Coaxial Cable	
• 10Base2 (ThinWire Ethernet)	10 Mbps
• 10Base5 (ThickWire Ethernet)	10 Mbps
Fiber-Optic Cable	
• 10Base-F (Ethernet)	10 Mbps
• 100Base-FX (Fast Ethernet)	100 Mbps
• FDDI (Fiber Distributed Data Interface) token ring	100 Mbps
• Gigabit Ethernet	1 Gbps
• 10-Gigabit Ethernet	10 Gbps
• 40-Gigabit Ethernet	40 Gbps
• 100-Gigabit Ethernet	100 Gbps

Twisted Pair (Ethernet wire cable)

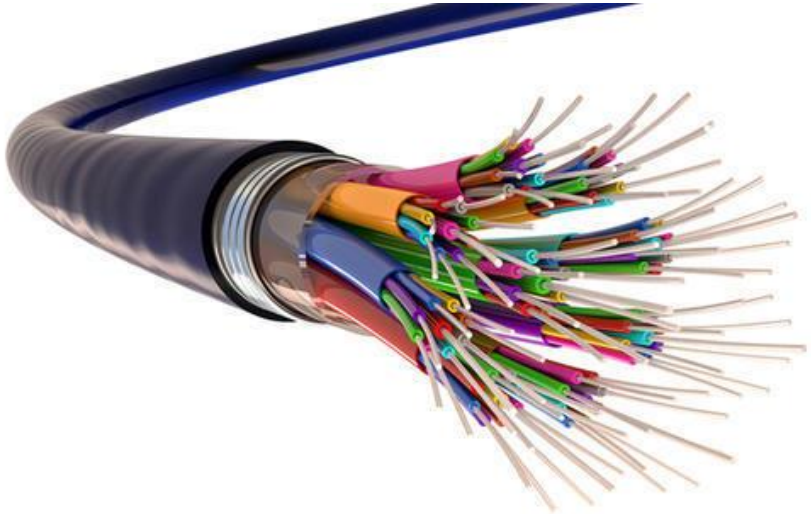


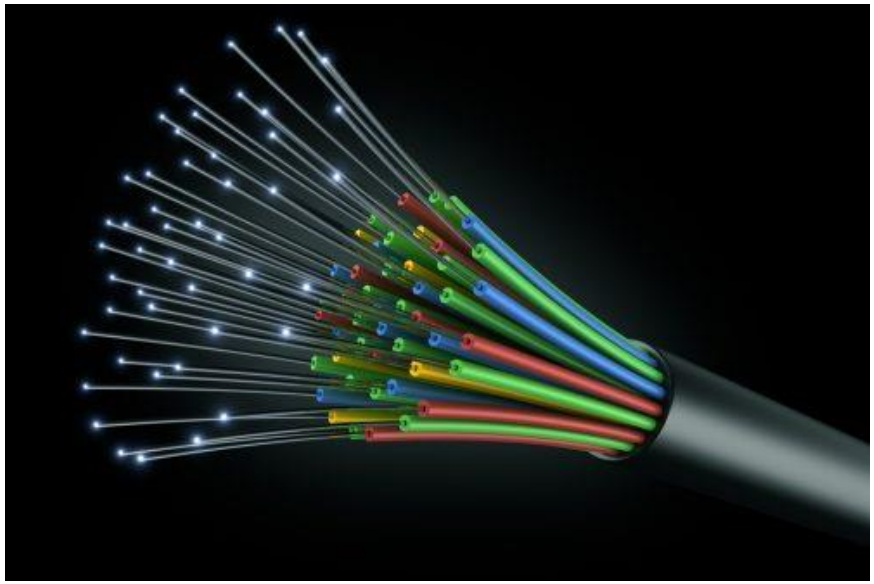


Coaxial Cable

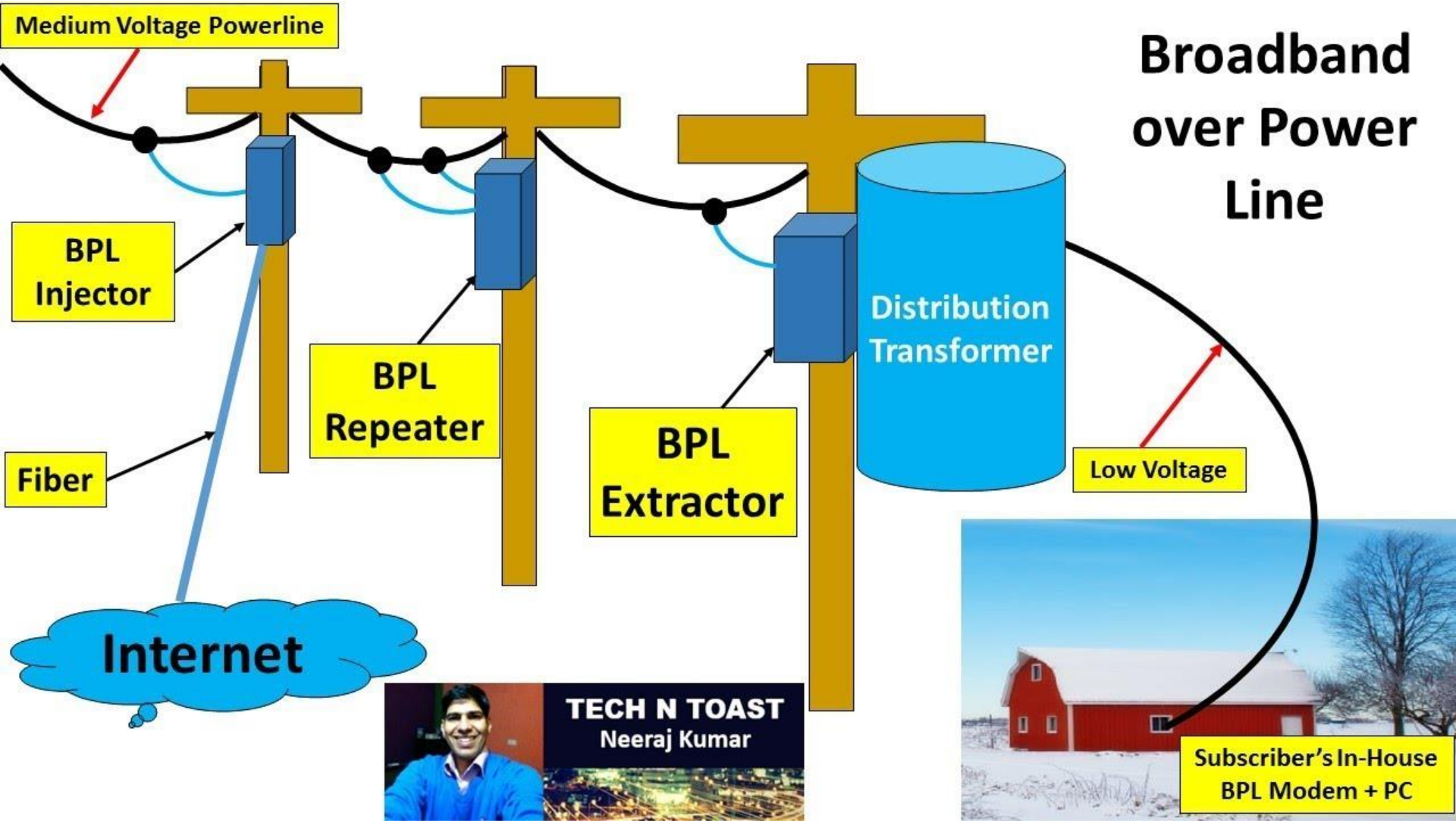


Fiber Optics Cable





Broadband over power line



Wireless Transmission Media

- Wireless transmission media send communications signals through the air or space.

- Many users opt for wireless transmission media because it is more convenient than installing cables.

Table 10-4 Wireless Transmission Media Transfer Rates

Medium		Maximum Transfer Transmission Rate
Infrared		115 Kbps to 4 Mbps
Broadcast radio	• Bluetooth	1 Mbps to 24 Mbps
	• 802.11b	11 Mbps
	• 802.11a	54 Mbps
	• 802.11g	54 Mbps
	• 802.11n	300 Mbps
	• 802.11ac	500 Mbps to 1 Gbps
	• 802.11ad	up to 7 Gbps
	• UWB	110 Mbps to 480 Mbps
Cellular radio	• 2G	9.6 Kbps to 144 Kbps
	• 3G	144 Kbps to 3.84 Mbps
	• 4G	Up to 100 Mbps
Microwave radio		10 Gbps
Communications satellite		2.56 Tbps

Components of computer network

1. Server
2. Transmission media
3. Network Interface Card
4. Switch/Hub
5. Router
6. PC/Client/work station

Types of Network

- A **local area network** (**LAN**) is a network that connects computers and devices in a limited geographical area such as a home, school, office building.
- **A *metropolitan area network* (*MAN*)** is a high-speed network that connects local area networks in a metropolitan area, such as a city or town, and handles the bulk of communications activity across that region.

Types of Network

- A **wide area network** (**WAN**) is a network that covers a large geographic area (such as a city, country, or the world) using a variety of wired and wireless transmission media.
- A **personal area network** (**PAN**) is a network that connects computers and devices in an individual's workspace using wired and wireless technology.
- Devices include smartphones, digital cameras, printers, and more.

Phone Hotspot (PAN)



INTERNET Vs. INTRANET

INTERNET	INTRANET
INTERnational **NET**work	**INT**ernal **R**estricted **A**ccess **NET**work
Information about everything	Information concerning a particular organisation that has set it up
Can be used by anyone	Used by employees, stake holders of the organization
Difficult to restrict the access	Certain websites outside the internal network can be blocked/restricted
Less secured	Secure from hackers as they are usually behind the firewall
No authentication necessary	Authentication required for access
Can be accessed from anywhere	Can be accessed from agreed points

Intranet Services

- Electronic mails
- Discussion groups/ Newsgroups
- Chatting and Instant messaging
- Internet Telephony (VoIP)
- File Transfer Protocol (FTP)
- Video conferencing
- World wide web (Websites)

Internet vs Web

- The Internet contains physical components
  - computers
  - networks
  - services
- The Internet connects thousands of computers across the world, but it is the web that allows communication to occur
- Web - abstraction and common set of services on top of the Internet
- Web - set of protocols and tools that let us share information with each other

Evolution of the Internet

- The Internet originated as ARPANET in September 1969 and had two main goals:

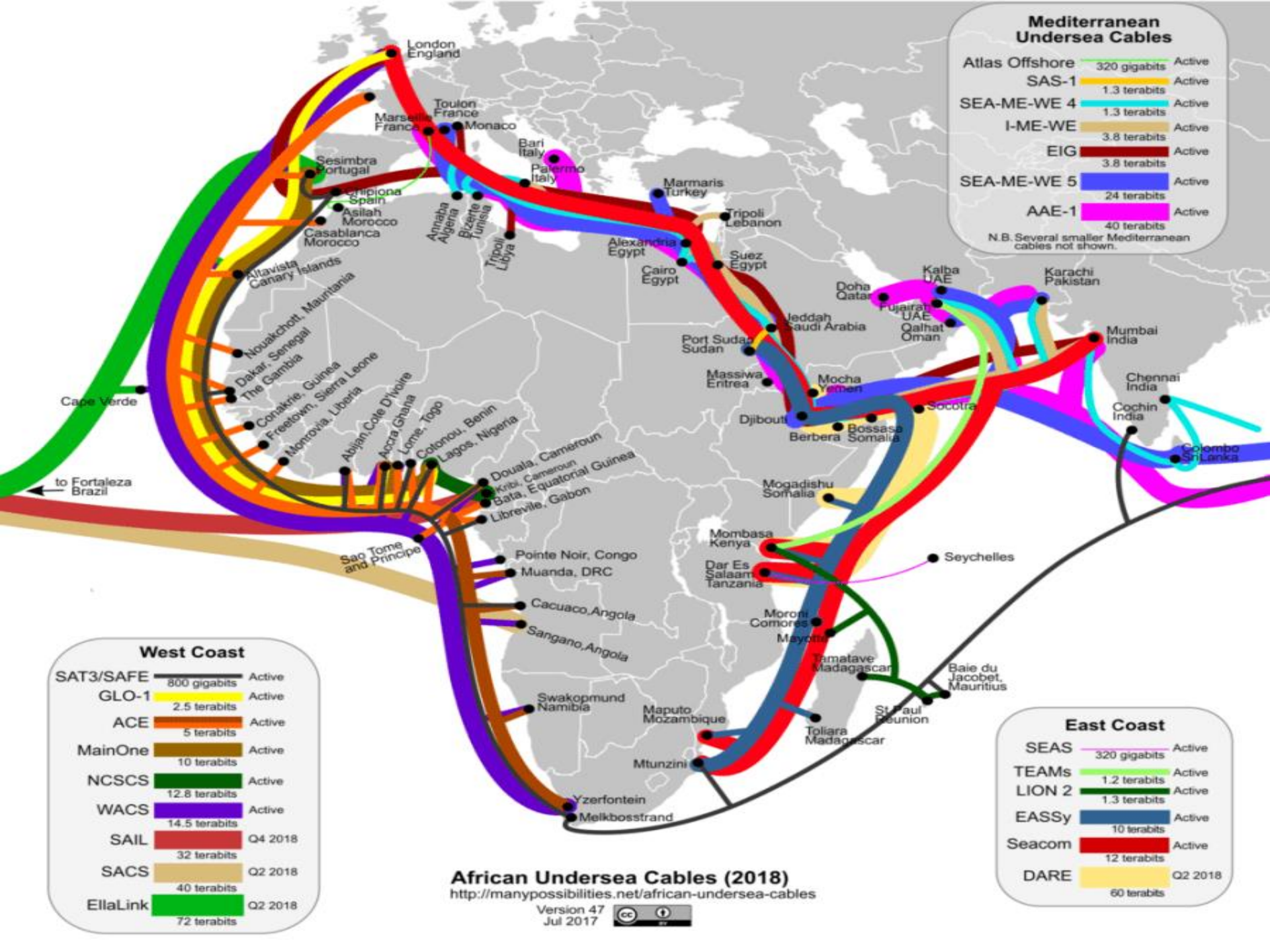
Allow scientists at different physical locations to share information and work together

Function even if part of the network were disabled or destroyed by a disaster

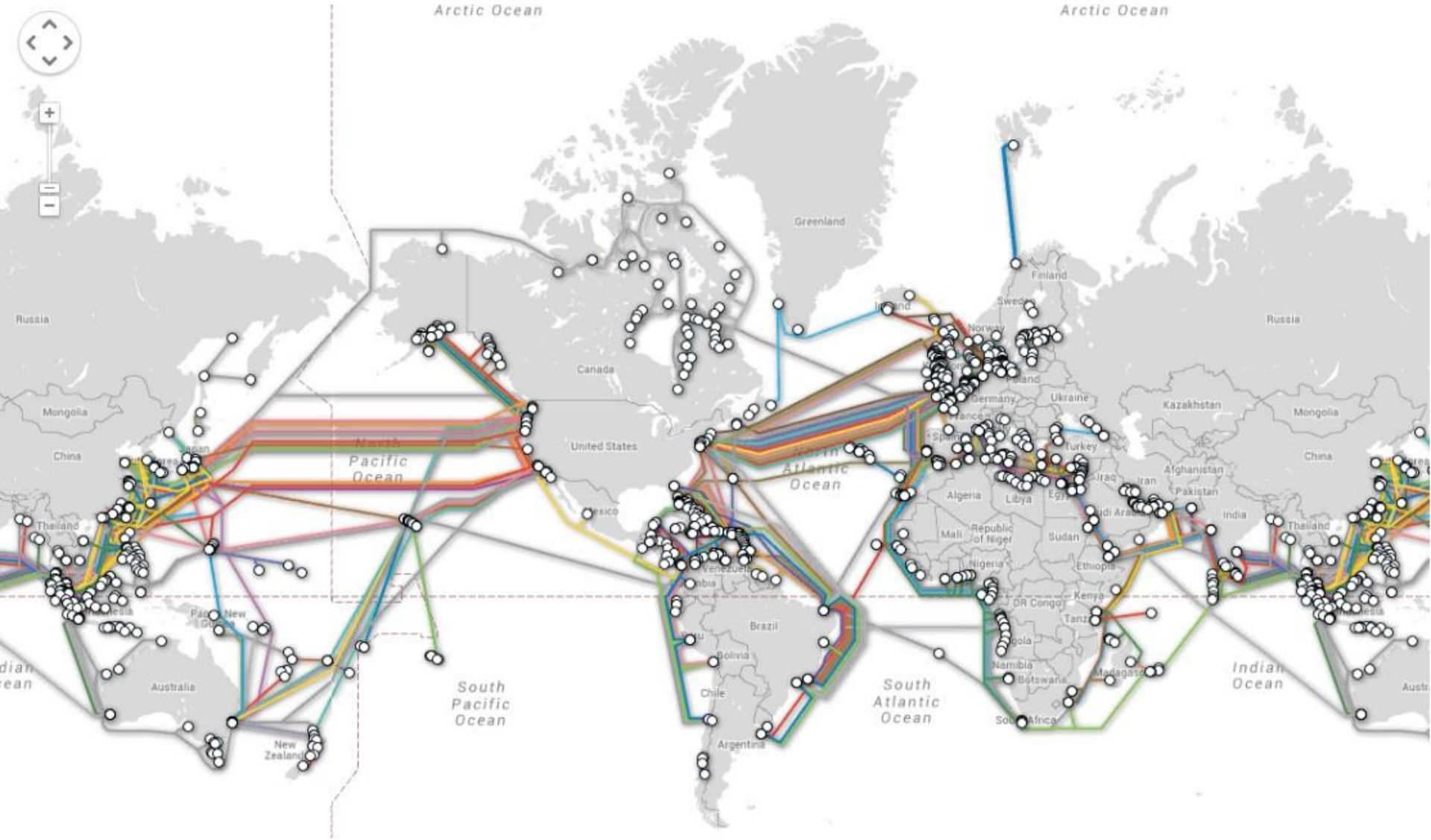
Evolution of the Web

- Sir Tim Berners-Lee is famously credited for the invention of the World Wide World in the early 1989.

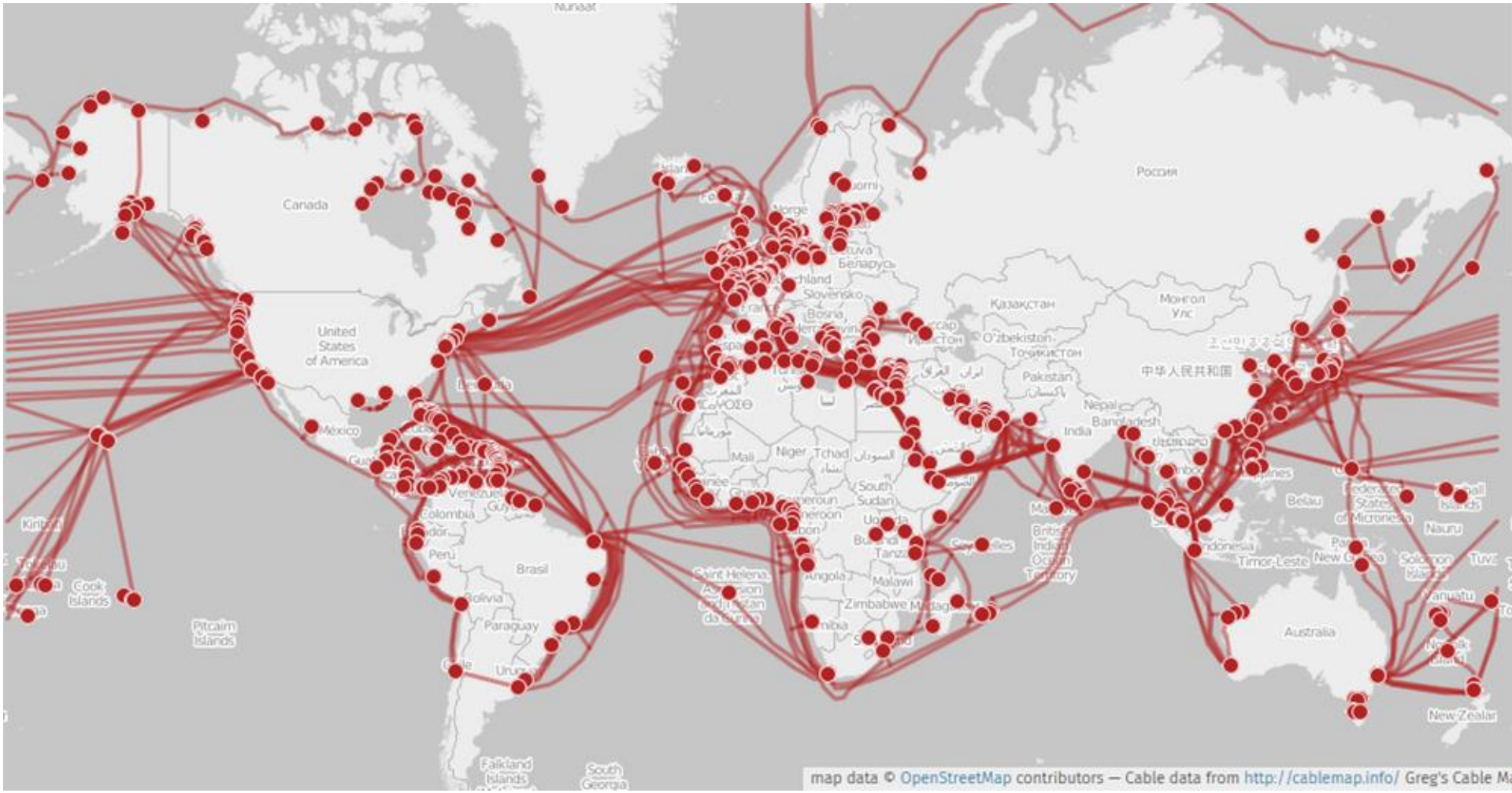



Mediterranean Undersea Cables
Atlas Offshore 320 gigabits Active
SAS-1 1.3 terabits Active
SEA-ME-WE 4 1.3 terabits Active
I-ME-WE 3.8 terabits Active
EIG 3.8 terabits Active
SEA-ME-WE 5 24 terabits Active
AAE-1 40 terabits Active
N.B. Several smaller Mediterranean cables not shown.
West Coast
SAT3/SAFE 800 gigabits Active
GLO-1 2.5 terabits Active
ACE 5 terabits Active
MainOne 10 terabits Active
NCSCS 12.8 terabits Active
WACS 14.5 terabits Active
SAIL 32 terabits Q4 2018
SACS 40 terabits Q2 2018
EllaLink 72 terabits Q2 2018
East Coast
SEAS 320 gigabits Active
TEAMs 1.2 terabits Active
LION 2 1.3 terabits Active
EASSy 10 terabits Active
Seacom 12 terabits Active
DARE 60 terabits Q2 2018
London England
Toulon France
Marseille France
Monaco
Bari Italy
Palermo Italy
Sesimbra Portugal
Chipiona Spain
Asilah Morocco
Casablanca Morocco
Annaba Algeria
Bizerte Tunisia
Tripoli Libya
Marmaris Turkey
Tripoli Lebanon
Alexandria Egypt
Cairo Egypt
Suez Egypt
Altavista Canary Islands
Nouakchott, Mauritania
Dakar, Senegal
The Gambia
Conakrie, Guinea
Freetown, Sierra Leone
Monrovia, Liberia
Abijan,Cote D'ivoire
Accra,Ghana
Lome, Togo
Cotonou, Benin
Lagos, Nigeria
Douala, Cameroun
Kribi, Cameroun
Bata, Equatorial Guinea
Libreville, Gabon
Cape Verde
to Fortaleza Brazil
Sao Tome and Principe
Pointe Noir, Congo
Muanda, DRC
Cacuaco,Angola
Sangano,Angola
Swakopmund Namibia
Maputo Mozambique
Mtunzini
Yzerfontein
Melkbosstrand
Doha Qatar
Kalba UAE
Fujairah UAE
Qalhat Oman
Karachi Pakistan
Jeddah Saudi Arabia
Port Sudan Sudan
Massiwa Eritrea
Mocha Yemen
Djibouti
Berbera
Bossasa Somalia
Socotra
Mumbai India
Chennai India
Cochin India
Colombo Sri Lanka
Mogadishu Somalia
Mombasa Kenya
Dar Es Salaam Tanzania
Seychelles
Moroni Comores
Mayotte
Tamatave Madagascar
St Paul Reunion
Baie du Jacobet, Mauritius
Toliara Madagascar
African Undersea Cables (2018)
http://manypossibilities.net/african-undersea-cables
Version 47
Jul 2017

Undersea cables that keep the Internet alive



Undersea cables that keep the Internet alive



Exploring The Hidden Internet



	Surface Web	Deep Web	Dark Web
Description	Content that search engine can find	Content that search engine cannot find	Content that are hidden intentionally
Known as	Visible web, Indexed web	Invisible web, Hidden web, Deep web	-
Constitutes	Web	Web	Web
Contents	Legal	Legal + Illegal	Illegal
Information found	4%	96%	-
Browsers	Google chrome, Mozilla firefox, opera	-	TOR browser

Connection to Internet

- Mobile broadband - USB modem (dongle)/MiFi
- Home Broadband - Digital Subscriber Line (DSL)
- WiFi Hotspot
- Cellular networks

Mobile Broadband

- Wireless Internet access through a portable modem.



Cellular network technology

Symbol	Name	Generation	Download Speeds
G	GPRS	2G	0.1 Mb/s
E	EDGE	2.5G	0.4 Mb/s
3G	3G	3G	2 Mb/s
H	HSDPA	3.5G	7.2 Mb/s
H+	HSDPA+	3.75G	21 Mb/s
LTE	LTE / 4G	4G	100 Mb/s

The promise of 5G network

1G	2G	3G	4G	5G
1981	1992	2001	2010	2020(?)
2 Kbps	64 Kbps	2 Mbps	100 Mbps	10 Gbps
Basic voice service using analog protocols	Designed primarily for voice using the digital standards (GSM/CDMA)	First mobile broadband utilizing IP protocols (WCDMA / CDMA2000)	True mobile broadband on a unified standard (LTE)	'Tactile Internet' with service-aware devices and fiber-like speeds
				?

Surfline 4G Devices



Surfline Modem

GHc99



Surfline Mi-Fi

GHc250



Surfline Router

GHc499

Surfline


POWER
WLAN
WPS
LAN
VOICE
surfline
4G
Surfline
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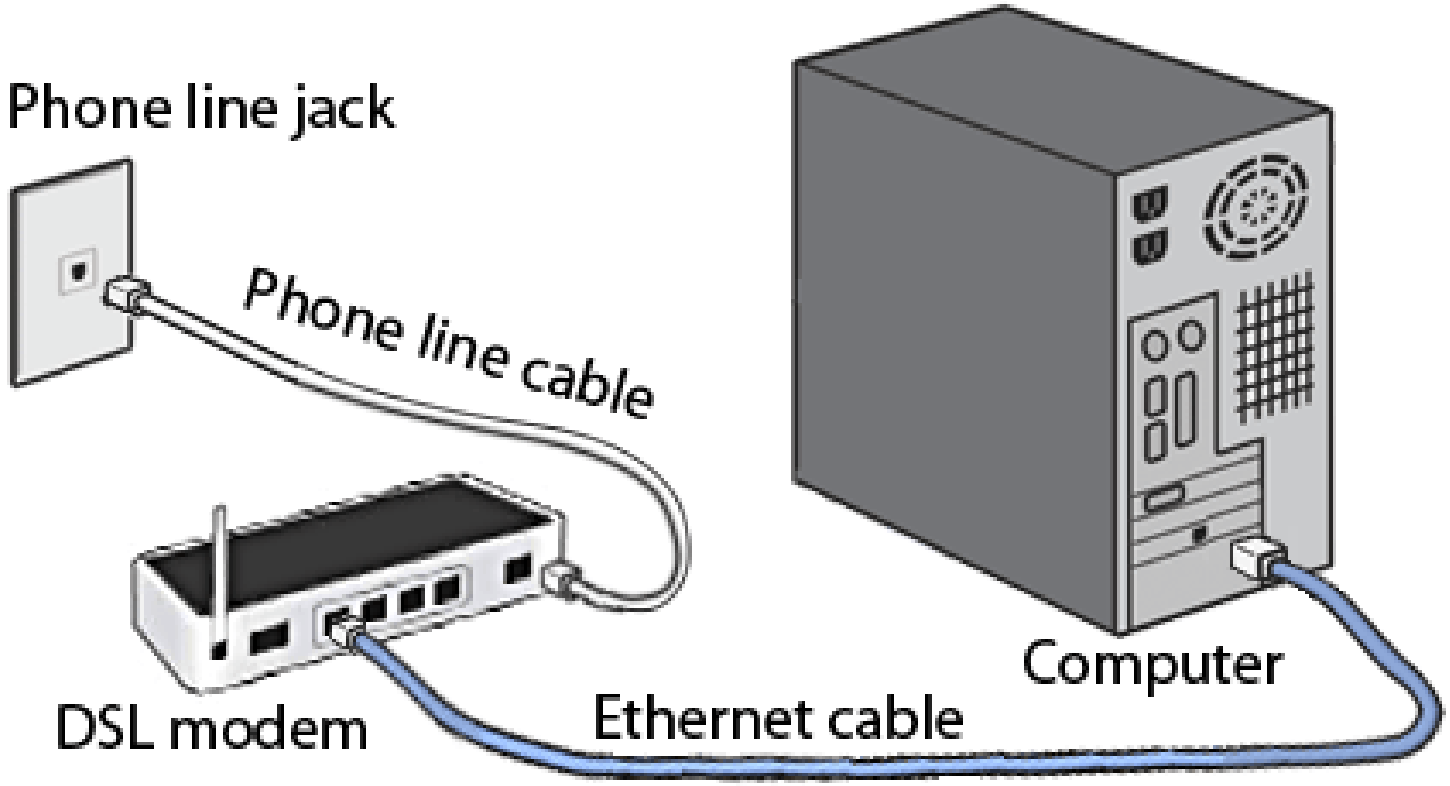
Home broadband

- Digital Subscriber Line (DSL)
  - Faster than mobile broadband
  - Requires a fixed telephone line





Digital Subscriber Lines



Cost of data - Vodafone

20GB	40GB	80GB
Browser	Streamer	Webmaster
GHS85	GHS130	GHS190
Show more	Show more	Show more

80GB	200GB	500GB
Webmaster	Downloader	Office
GHS190	GHS265	GHS420
Show more	Show more	Show more

Connecting to the Internet

- An **Internet service provider** (**ISP**) is a business that provides individuals and organizations access to the Internet free or for a fee
- Bandwidth represents the amount of data that travels over a network
  - Megabyte (MB)
  - Gigabyte (GB)

Internet Service Providers

- List of companies:
  - All telcos
  - Busy internet
  - Zipnet
  - iBurst
  - Main one
  - Internet Ghana
  - Glo-1
  - Surline

Connecting to the Web

- Web browser is software that allows you to locate, view, and navigate the Web.



Getting Around the Web

- Web sites
- URLs
- Hyperlinks
- Favorites and Bookmarks



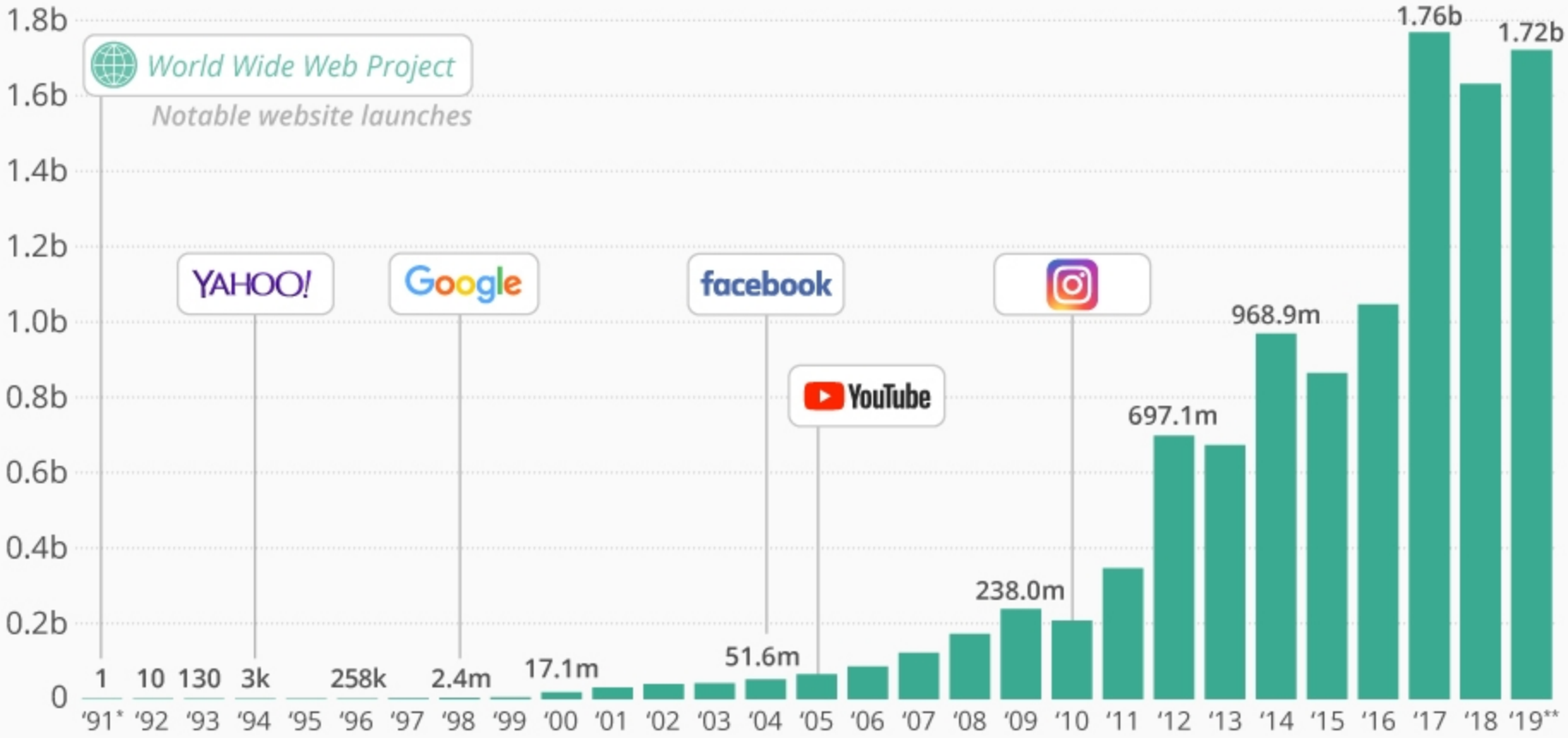
Web Sites

- Web site:
  - Collection of related Web pages
  - First page known as Home or Index page
- Web page:
  - HTML document
    - Text and graphics
  - Unique address
  - Hyperlinks



How Many Websites Are There?

Number of websites online from 1991 to 2019



"Website" is defined as a unique hostname, i.e. a name which can be resolved, using a name server, into an IP Address.

* As of August 1, 1991
** As of October 28, 2019 at 10:00 CET

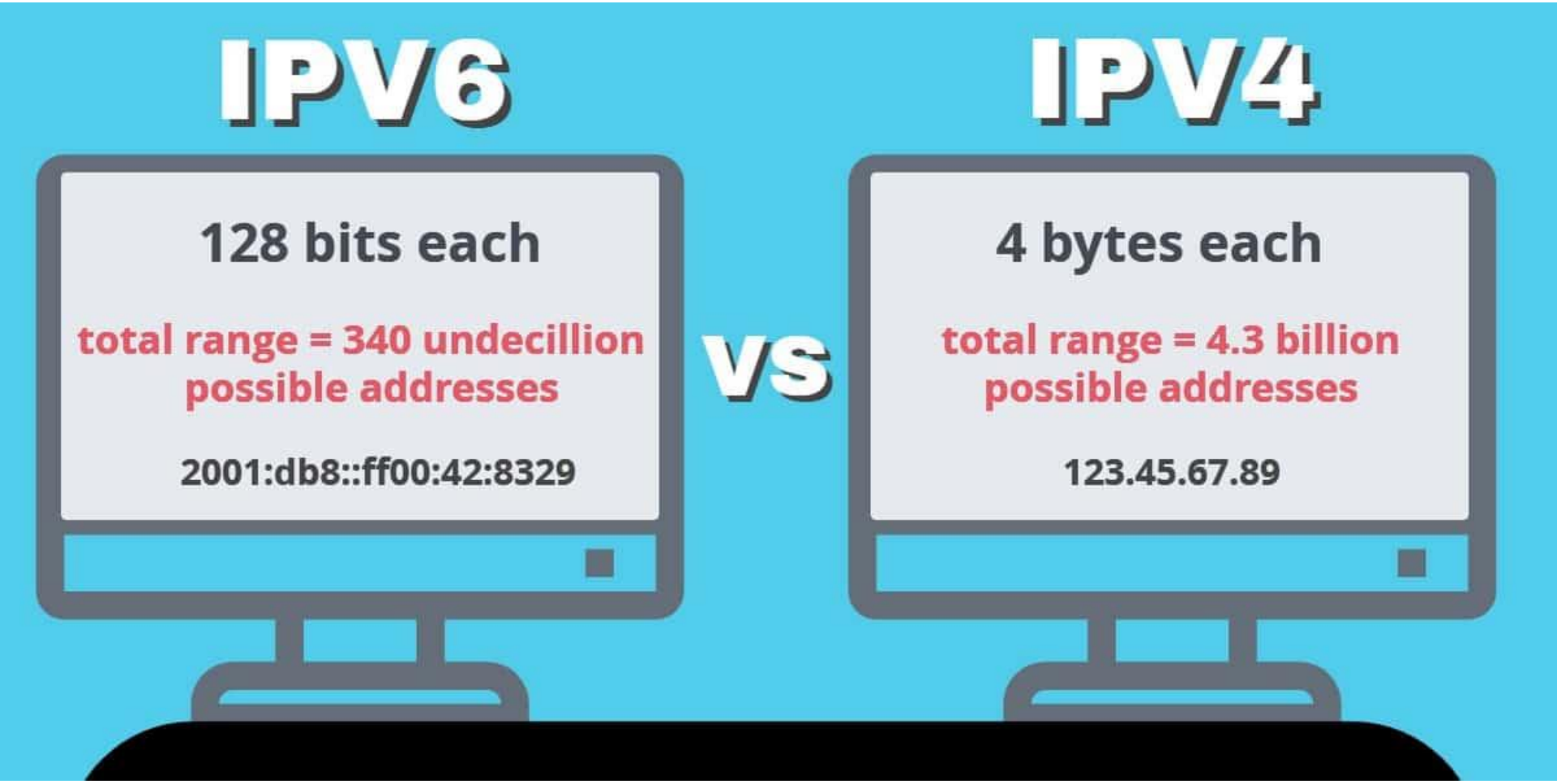
@StatistaCharts

Source: Internet Live Stats

statista

Internet Addressing and Architecture

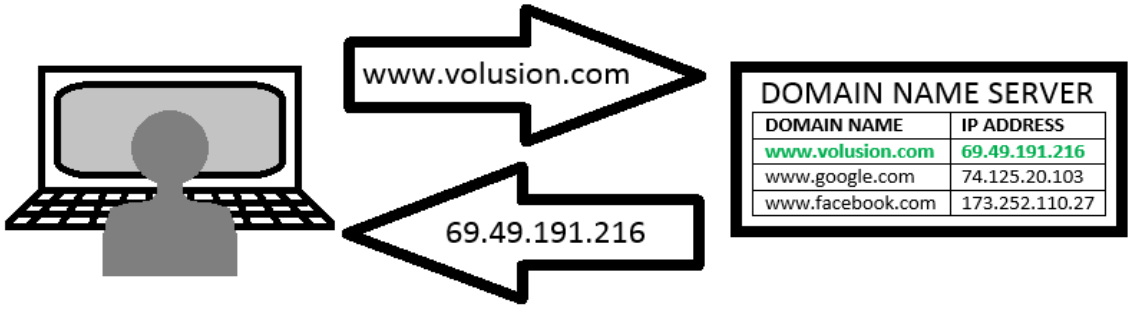
- Internet Protocol (IP) Address
- Every computer on the Internet is assigned a unique IP.
- To access a computer on the internet, you only have to specify its IP address.
- An example of IP address: 207.46.250.119, 10.0.0.1
- The numbers ranges from 0 – 255


IPV6
128 bits each
total range = 340 undecillion possible addresses
2001:db8::ff00:42:8329
VS
IPV4
4 bytes each
total range = 4.3 billion possible addresses
123.45.67.89

	Internet Protocol version 4 (IPv4)	Internet Protocol version 6 (IPv6)
Deployed	1981	1999
Address Size	32-bit number	128-bit number
Address Format	Dotted Decimal Notation: 192.149.252.76	Hexadecimal Notation: 3FFE:F200:0234:AB00: 0123:4567:8901:ABCD
Prefix Notation	192.149.0.0/24	3FFE:F200:0234::/48
Number of Addresses	2^{32} = ~4,294,967,296	2^{128} = ~340,282,366, 920,938,463,463,374, 607,431,768,211,456

Internet Addressing and Architecture

- Domain Name and Domain Name Systems
- Remembering the IP address is difficult hence an english-like description of IP hence **domain names.**
- Domain Name is the English-like name that corresponds to an IP address.
- **Domain Name system (DNS)** converts domain names to IP addresses.



Internet architecture and governance

- No formal management: **IAB, ICANN, W3C**
- **IAB** - Internet Architecture Board
  - Internet Society that overseer the technical evolution of the Internet.
- **ICANN** - Internet Corporation for Assigned Names and Numbers
  - a private (non-government) non-profit corporation with responsibility for IP address space allocation, protocol parameter assignment, domain name system management etc.
- **W3C -** World Wide Web Consortium
  - develop standards for the World Wide Web.

List of countries by IPv4 address allocation



Rank	Country or entity	IP addresses[3]	%	Population (mostly 2012)[4]	IP addresses per 1000
	World	4,294,967,296	100.0	7,021,836,029	611
1	United States	1,541,605,760	35.9	313,847,465	4,911
	Bogons	875,310,464	20.4		
2	People's Republic of China	330,321,408	7.7	1,343,239,923	245
3	Japan	202,183,168	4.7	127,368,088	1,587
4	United Kingdom	123,500,144	2.9	63,047,162	1,958
5	Germany	118,132,104	2.8	81,305,856	1,452
6	South Korea	112,239,104	2.6	48,860,500	2,297
7	France	95,078,032	2.2	65,630,692	1,448
8	Canada	79,989,760	1.9	34,300,083	2,332
9	Italy	50,999,712	1.2	61,261,254	832
10	Brazil	48,572,160	1.1	205,716,890	236
99	Paraguay	365,312	0.0	6,541,591	55.84
100	Mozambique	343,296	0.0	23,515,934	14.60
101	Tanzania	339,712	0.0	43,601,796	7.79
102	Ghana	332,544	0.0	25,241,998	13.17
	Macau	324,864	0.0	578,025	562.02
103	Albania	323,840	0.0	3,002,859	107.84

Countries can block websites

IGP considers blocking social media on election day

Source: Ghana|myjoyonline.com|EA

Date: 26-05-2016 **Time:** 07:05:46:pm

UN opposed to social media ban in Ghana – Ibn Chambas

Source: Ghana | Myjoyonline.com | Austin Brako-Powers | Email: Austin.powers@myjoyonline.com

Date: 18-06-2016 **Time:** 03:06:03:pm

Ghana: Social media blackout on election day debate deepens

General News of Monday, 30 May 2016 **Source:** globalvoices.org

Ghanaians oppose shutdown of social media during elections

Which websites are blocked in China?

Here's a list of the most popular websites which, at the moment, are blocked in China:

Social

Facebook, Twitter, Instagram, Pinterest, Tumblr, Snapchat, Picasa, WordPress.com, Blogspot, Blogger, Flickr, SoundCloud, Google+, Google Hangouts, Hootsuite.

Apps

Google Play (this means that without a VPN you won't be able to download any apps from Google Play), Whatsapp, Messenger, Viber, Telegram, Line, Periscope, KaKao Talk, Discord.

Search engines

Google (both Google.com and the majority of local versions like Google.com.hk, Google.fr, etcetera), Ask.com, Duck Duck Go, Yahoo, various foreign versions of Baidu.

Video Sharing

Youtube, Vimeo, Daily Motion, Nico Video.

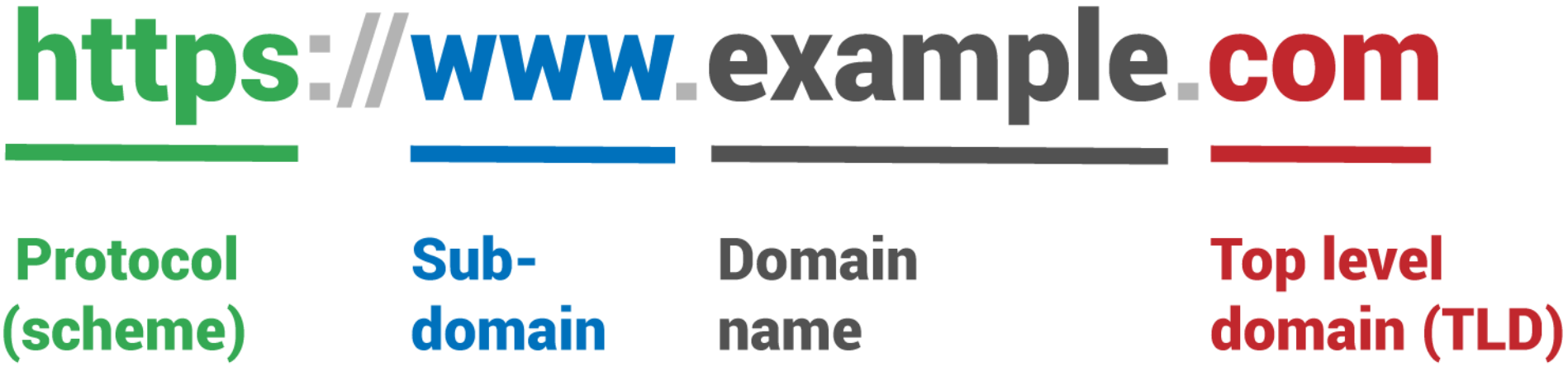
Media

The New York Times, Finacial Times, The Wall Street Journal, The Economist, Bloomberg, Reuters, The Independent, LeMonde, L'Equipe, Google News, many pages of Wikipedia, Wikileaks.

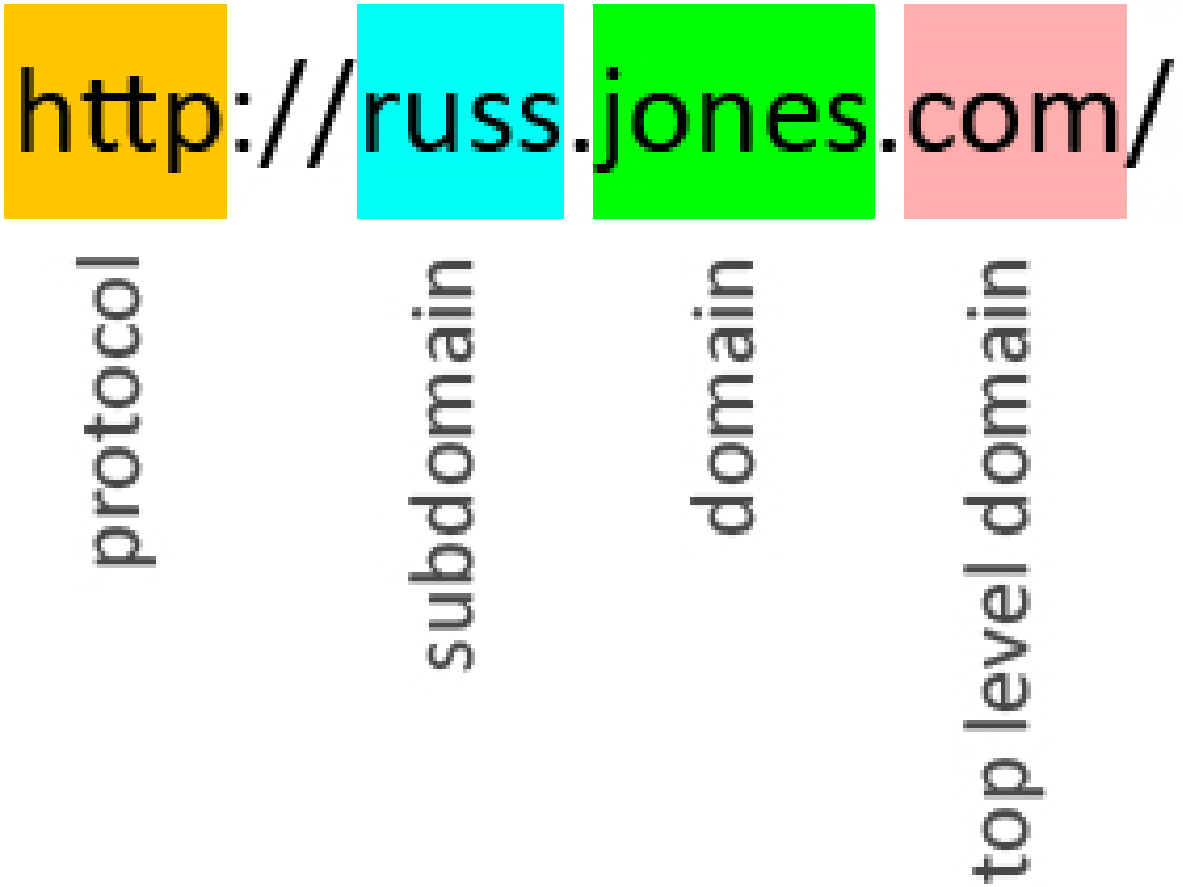
Uniform Resource Locator (URL)

- URL is the address of a web page or the address of a resource on the Internet.
- Example URLs:
  - www.ucc.edu.gh
  - https://snm.ucc.edu.gh/
  - https://nursing.ucc.edu.gh/
  - www.ghana.gov.gh
  - www.ges.gov.gh
  - www.vodafone.com.gh
  - www.google.com
  - www.google.com.gh

Parts of web address or URL



URL PARTS


http://russ.jones.com/
protocol
subdomain
domain
top level domain

URL of MOH

- Identify the paths of this MOH URL:

- **https://portal.ucc.edu.gh**

Current Top-Level Domains

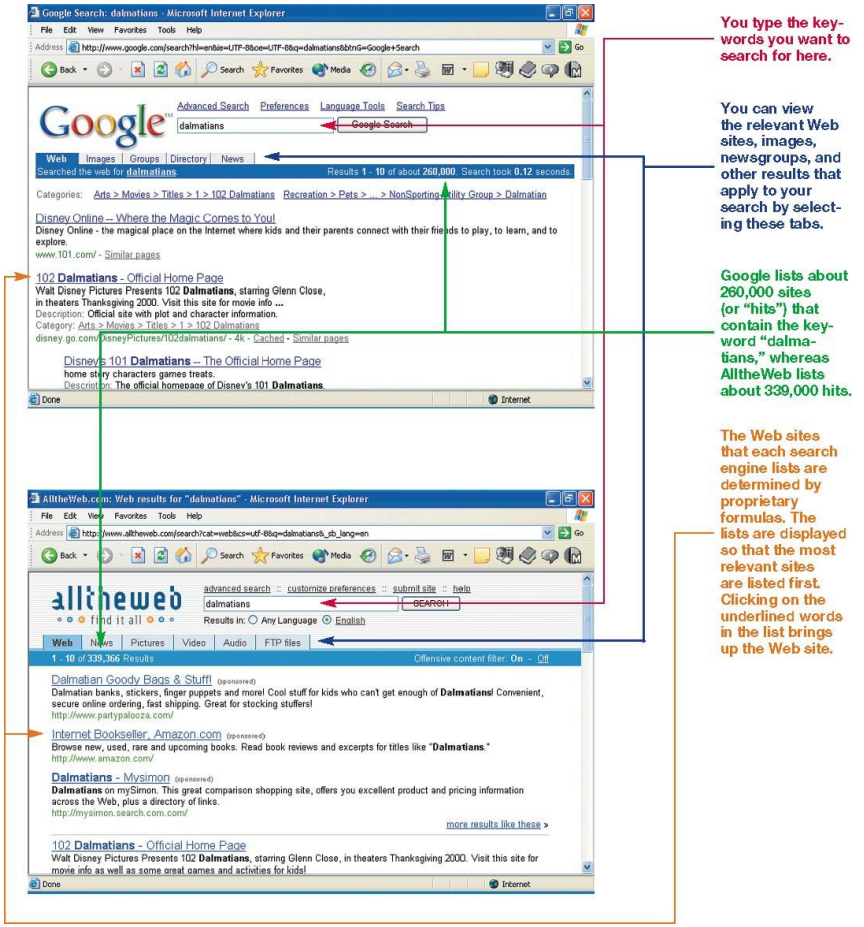
TLD	Meaning
.com	Can be used by anyone (commercial)
.edu	Academic institution
.gov	Governments
.info	Information Service Providers
.net	Networking Organisation
.org	Organisations (often non-profit)
.gh .com.gh or .edu.gh	Country domain

There are 13 basic types of websites

1. *Advocacy*
2. *News*
3. *Portal*
4. *Personal*
5. *Marketing/business*
6. *Blog*
7. *Online social network*
8. *Entertainment*
9. *Wiki*
10. *Informational*
11. *Educational*
12. *Content aggregator*
13. *Web application*

Search Engines

- User keys word or phrase in search box
- “Spider” or “Web Crawler” program scans web pages
- Results are indexed and sent to the client



Criteria for Evaluating the Accuracy/Validity of a Website

- **Authority and Affiliation**
  - Authority reveals that the person, institution or agency responsible for a site has the qualifications and knowledge to do so.
  - Contact information should be clearly provided: e-mail address, snail mail address, phone number, and fax number.
  - Credentials: the author should state qualifications, credentials, or personal background that gives them authority to present information.
  - Check to see if the site supported by an organization or a commercial body.

Criteria for Evaluating the Accuracy/Validity of a Website

- **Audience**
  - Who is the website's intended audience?

    Academics? The general public? School children?
  - Does it appropriately address the target audience?
  - Is it relevant for your assignment or research?

Criteria for Evaluating the Accuracy/Validity of a Website

- **Accuracy**
  - Is it free from spelling errors?
  - Is the text well-written and grammatically correct?
  - Has the content been through an editing process or been peer reviewed?
  - Has the author included a bibliography?
  - Are the sources cited reliable and can they be verified elsewhere?
  - Are research methodologies adequately explained?

Criteria for Evaluating the Accuracy/Validity of a Website

- **Objectivity**
  - Does the author present objective arguments or make it clear when he is expressing biased opinions?
  - Are other points of view explored?
  - Is it a personal website? Does it express personal opinions?
  - Is the website part of a commercial organization, a political party or an organization with a specific agenda? If yes, question the motives for publishing the information.
  - Does the website promote a biased viewpoint?

Criteria for Evaluating the Accuracy/Validity of a Website

- **Currency**
  - Can you tell when the information was published?
  - Is the information up to date?
  - How frequently is the website updated?
  - Are the links up to date and working?

E-commerce

- Three major types of e-commerce
  - Business-to-consumer (B2C)
    - Example: amazon.com, Aliexpress.com, KiKuu.com, jumia.com.gh
  - Business-to-business (B2B)
    - Example: Alibaba.com
  - Consumer-to-consumer (C2C)
    - Example: eBay, tonaton, olx

- E-commerce can be categorized by platform
  - Mobile commerce (m-commerce)

E-commerce revenue models

- Advertising
- Sales
- Subscription
- Free/Freemium
- Transaction fee
- Affiliate

INSTA KING Cristiano Ronaldo earns £37MILLION from Instagram – more than he does playing for Juventus

REVEALED

Tatiana Sanchez
18 Oct 2019, 19:19 | Updated: 18 Oct 2019, 20:23

Instagram handle: @cristiano

Niche: Celebrity, sports

Followers: 142 million

Cost per-post: $750,000

How to generate revenue from the web

- How to generate revenue from youtube.
- Enable AdSense
- Your youtube videos with 1000 views you earn **$7.6**
- **Gangnam Style** had 1 billion views and was paid **$7.8 million**

Home › Entertainment › Ghanaian musicians paid YouTube royalties

Entertainment

Ghanaian musicians paid YouTube royalties

By **ameyawdebrah.com** - Jul 31, 2013 282



Ohene Media pays YouTube royalties to Ghanaian musicians

Content Management System (CMS)

- CMS is software that helps users create, manage, and modify content on a website without the need for specialized technical knowledge.
- In simpler language, a CMS is a tool that helps you build a website without needing to write all the code from scratch (or even know how to code at all).

Popular content management systems

- WordPress
- Joomla
- Drupal
- Magento (for eCommerce stores)
- Squarespace
- Wix
- ExpressionEngine
- Alfresco

E-commerce apps







Mobile Payment Apps

- Bank Apps
- Mobile Money apps
- Third party apps
  - Expresspay
  - Slydepay
  - Zeepay
  - Hubtel
  - Mazzuma
  - Myghpay
  - Interpay
  - Paypal*

Netiquette

- **Netiquette**, which is short for Internet etiquette, is the code of acceptable behaviors users should follow while on the Internet; that is, it is the conduct expected of individuals while online.

Netiquette Guidelines for Online Communications

Golden Rule: Treat others as you would like them to treat you.

Be polite. Avoid offensive language.

Avoid sending or posting *flames*, which are abusive or insulting messages. Do not participate in *flame wars*, which are exchanges of flames.

Be careful when using sarcasm and humor, as it might be misinterpreted.

Do not use all capital letters, which is the equivalent of SHOUTING!

Use **emoticons** to express emotion. Popular emoticons include:

:) Smile :| Indifference :o Surprised :(Frown :\ Undecided ;) Wink

Use abbreviations and acronyms for phrases:

BTW	by the way	IMHO	in my humble opinion	FWIW	for what it's worth
FYI	for your information	TTFN	ta ta for now	TYVM	thank you very much

Clearly identify a spoiler, which is a message that reveals an outcome to a game or ending to a movie or program.

Be forgiving of other's mistakes.

Read the *FAQ* (frequently asked questions), if one exists.